

Part 3. Written Insights

RFM Scoring and Segmentation Approach

RFM is a commonly used technique that evaluates customer value and segments customers in a way that can be used to build marketing strategies.

The metrics that I used in order to rank customers are three:

- **Recency**: how recently a customer bought
- **Frequency**: how often a customer buys
- **Monetary**: how much a customer spends

To create the ranks I used the `NTILE(5)` function, which splits customers into 5 equally sized groups where 1 is the lowest and 5 the highest. The top 20% get a score of 5 and the lowest 20% get a score of 1. The `OVER (ORDER BY ...)` clause is what tells `NTILE` which direction to rank in, and getting that direction right turned out to be the most important detail in the whole model.

Because `NTILE` assigns bucket 1 to whatever sorts first, Recency and the other two metrics have to be sorted in opposite directions. Recency is stored as days since the last order, so a bigger number is worse and the sort runs descending. Frequency and Monetary run the other way, where bigger is better, so those sort ascending. I also added `customer_id` as a tiebreaker, because without it customers with identical values were assigned to different buckets on different runs and the model was not reproducible.

Finally I used a `CASE` statement to turn the scores into readable groups such as 'Champions' or 'Need Attention'. Since `CASE` returns the first branch that matches, the conditions have to run from most specific to most general. A broad condition placed above a narrow one makes the narrow one unreachable, which is what happened to my 'Lost' branch in the first version of this model: it sat below `r_score <= 2` and never returned a single customer.

A note on the Monetary dimension. I originally scored Monetary on total spend. During the analysis I found that total spend and frequency correlate at **0.94** in this dataset, meaning they were measuring nearly the same thing and the model was effectively RF rather than RFM. Average order value, by contrast, correlates with frequency at **-0.008**, so it carries information that total spend was hiding. I therefore scored Monetary on average order value instead. The effect is described in the next sections.

What the Monetary change did

With Monetary scored on total spend, the 'Champions' segment held about 19,500 customers — 19.5% of the base. If Recency, Frequency and Monetary were independent, three quintile filters at score 4 or above should select roughly 6% of customers. Getting three times that was the first sign that two of the three conditions were redundant.

After switching Monetary to average order value, 'Champions' fell to 8,136 customers, or 8.14%, which is close to what independence predicts. The top five customers by RFM score also changed sharply: their total spend rose from around 3.19M each to between 5.87M and 6.06M. The model is now selecting customers who both order often and spend well per order, instead of selecting the same "orders often" group twice.

Key RFM Driver

I originally concluded that Recency was the most significant driver of customer behaviour. Looking at the model more carefully, I cannot support that claim from these results. Every branch of my `CASE` statement tests `r_score` first, so Recency was guaranteed to dominate the segmentation by design. The output reflects how I built the model rather than what the data says.

What the data does show is that Frequency and Monetary were not independent. In this dataset, total spend is almost entirely a function of how many orders a customer places — average order value sits near 150,000 across all five Monetary quintiles, so spending more overall mostly means ordering more often, not ordering bigger.

Testing which dimension genuinely separates customer value would require scoring each one on its own and comparing revenue per customer across the resulting groups. That is a reasonable next step rather than something this model already answers.

Segment Distribution Insights

Segment	Customers	% of customers	% of revenue	Revenue per customer
At risk	31,543	31.54%	29.54%	2,124,595
Loyal	21,367	21.37%	25.73%	2,731,975
Need Attention	13,263	13.26%	16.98%	2,904,770
Champions	8,136	8.14%	13.36%	3,725,194
Potential Loyalist	10,495	10.50%	6.52%	1,408,536
Lost	8,456	8.46%	3.78%	1,013,203
Others	6,736	6.74%	4.08%	1,373,919

Reading customer share against revenue share on the same row is what makes this table useful. 'Champions' hold 8.14% of customers but 13.36% of revenue, so they earn about 1.6 times their weight. 'Lost' hold 8.46% of customers and only 3.78% of revenue, less than half their weight. Revenue per customer runs cleanly from 3.7M for 'Champions' down to 1.0M for 'Lost', a spread of roughly 3.7 times, which tells me the segmentation is separating customers by value rather than just labelling them.

'At risk' is the largest group in both customers and revenue. These are customers who have not purchased recently but who spent substantially while they were active — nearly 2.1M each on average. They are not low-value customers; they are dormant ones, and that distinction matters for what to do about them.

Strategic Recommendations

Champions and Loyal together are 29.5% of customers and 39.1% of revenue. Retention here protects the largest concentration of value in the business. A tailored recognition or loyalty programme is justified, and the cost of keeping them is almost certainly lower than the cost of replacing them.

At risk is where I would focus next, and this reverses my original position. I previously wrote that this segment was not worth spending money on. The data does not support that: 'At risk' holds 29.54% of total revenue at over 2.1M per customer. Writing them off means writing off close to a third of historical revenue. A reactivation campaign aimed at this group has more revenue behind it than any other single action available.

Need Attention sits at 2.9M revenue per customer, higher than 'Loyal', but with weaker recency. These customers are still valuable and have only recently started slipping. Acting here is cheaper than acting after they reach 'At risk'.

Potential Loyalist bought recently but infrequently, at 1.4M per customer. The opportunity is to increase order frequency rather than order size.

Lost is the one group where limited investment is defensible: lowest recency, lowest frequency, lowest revenue per customer, and the smallest revenue contribution of any segment.

Geographic Analysis

I originally reported that Romania had the most high-value customers, followed by Indonesia and Mozambique. That finding does not hold, for two reasons.

First, I ranked nations by raw count of 'Champions'. Since TPC-H distributes customers nearly evenly across 25 nations, that ranking mostly reflected nation size. Ranking by the share of each nation's customers who are Champions puts Romania 24th out of 25, not first.

Second, even the corrected ranking is not meaningful. Champion rates run from 9.11% (Algeria) down to 7.35% (Kenya), a spread of 1.76 percentage points. With about 4,000 customers per nation and an overall rate of 8.14%, one standard deviation is roughly 0.43 percentage points, and the expected gap between the highest and lowest of 25 groups is around 1.7 points from random variation alone. The observed spread is 1.76.

The conclusion is that Champion rates are statistically indistinguishable across all 25 nations. This is what I would expect from TPC-H, since the generator assigns customers to nations independently of their ordering behaviour.

Limitations

The dataset is synthetic. TPC-H generates orders pseudo-randomly, so several patterns here are properties of the generator rather than of customer behaviour. Average order value clusters tightly around 150,000, the top five customers

differ by only about 3%, and the near-zero correlation between frequency and average order value exists because order prices are generated independently of who is ordering. The method transfers to real data; these specific numbers do not.

A third of customers are not in the model. Every figure above covers only the 99,996 customers who placed at least one order. TPC-H's remaining 50,004 customers have no orders, therefore no recency, and cannot be scored. Arguably they are the most genuinely lost group in the dataset, and they sit outside every segment in this analysis.

RFM score treats the dimensions as interchangeable. Adding $r + f + m$ means a customer scoring 5-1-5 and one scoring 3-4-4 both total 11, despite describing very different behaviour. The composite score is useful for a rough ranking but should not be read as a single measure of value.

Quintiles are relative, not absolute. `NTILE(5)` always produces five equal groups, whether or not the underlying values differ meaningfully. A score of 5 means "top fifth of this dataset", not "high" in any absolute sense.

Further Analysis

Scoring each dimension separately and comparing revenue per customer across the resulting groups would test which of R, F and M actually drives value, rather than assuming it.

Tracking movement between segments over time would show whether retention efforts are working — in particular the flow from 'Need Attention' into 'At risk', which is the point where customers become expensive to recover.

Analysing which products each segment buys would allow targeted offers rather than one campaign per segment.

Finally, the 50,004 customers with no orders deserve their own analysis. Understanding why they registered and never purchased is a different question from retention, and potentially a more valuable one.